

The $n = 6$ Exclusivity Atlas

A verify-driven closed-negative cluster: how machine-checked falsification tightens the $\{1, 6\}$ arithmetic-function locus from an identity to an exclusive, non-lifting, scale-stable phenomenon

TECS-L domain (NOVEL axis, F family), **hexa-lang**
dancinlab

github.com/dancinlab/hexa-lang

May 26, 2026

Abstract

The first perfect number 6 sits at the centre of a small cluster of arithmetic-function identities — most prominently $\sigma(n)\phi(n) = n\tau(n)$, which a closed-form theorem (the TECS-L atlas atom M10) proves holds for *exactly* $n \in \{1, 6\}$ over all positive integers. A natural follow-up question is not whether the identities hold — that is settled — but whether the *specialness* of $\{1, 6\}$ is an isolated coincidence or whether it propagates: to amicable / quasi-perfect / multiply-perfect neighbours, to perfect-number siblings such as 28, to large or structurally notable arguments, and upward to the modular-curve tower $\Gamma_0(N) \rightarrow \Gamma_1(N) \rightarrow X(N)$. This paper aggregates the *NOVEL axis* of the TECS-L verification campaign — the lane that hunts *outside* the known, as opposed to the lanes that re-ground known identities — and shows that it operates as a **verify-driven exclusivity engine**. Rather than only confirming $n = 6$ identities, it systematically *rules out* the adjacent space: we pre-register and then machine-check (under the **hexa verify** tier rubric, §2) a battery of plausible “ $n = 6$ -like” generalisations and find them all *deterministically false*. Concretely we report (i) seven exact-integer closed-negatives from the closed-negative miner (F5); (ii) a non-lift result showing $\sigma\phi = n\tau \Leftrightarrow \{1, 6\}$ is a Γ_0 -level phenomenon that does *not* survive ascent to $\Gamma_1(N)/X(N)$, whose index is smooth in N (F7); and (iii) a scale-extension sweep confirming $D(n) := \sigma(n)\phi(n) - n\tau(n) \neq 0$ at seven notable arguments up to $n = 33550336$, extending the original $[1, 100]$ sweep by a factor of $\sim 3.4 \times 10^5$ (F6). Counting both the F5 falsifiers and the F6/F7 negatives, the cluster comprises *ten or more deterministic closed-negatives*. The headline is the *cluster itself*: $\{1, 6\}$ is not merely the support of an identity but an *exclusive, non-lifting, and scale-stable* arithmetic locus. Every negative is a pre-registered falsifier resolved by exact integer arithmetic — no tolerance, no probabilistic claim — which satisfies the `paper_significance` and `paper_negative_ok` gates of the host project’s discovery governance.

Contents

1	Statement: the exclusivity thesis and its falsifiers	2
1.1	Background: the positive kernel	2
1.2	The exclusivity thesis	2
2	Method: closed-negative mining under hexa verify	3
2.1	The verification discipline and tier rubric	3
2.2	The three mining lanes	4
2.3	Provenance and audit	4

3	Verification	4
3.1	F5: the seven closed-negatives (component E)	4
3.2	F6: scale-stability of $D(n) \neq 0$ (component S)	5
3.3	F7: non-lift to the modular-curve tower (component N)	6
4	Finding: the closed-negative cluster	7
4.1	Why a negative result is the finding	7
5	Caveats	8
6	Related work	9
A	F5 seven-falsifier verdict table (verbatim summary)	9
B	F6 notable-n table (verbatim sweep summary)	10
C	$\Gamma_1(N)/X(N)$ index table (verbatim assembly)	10
D	Raw verdict transcripts (verbatim ASCII)	10

1 Statement: the exclusivity thesis and its falsifiers

1.1 Background: the positive kernel

Let σ , ϕ , τ , μ denote the classical arithmetic functions (sum of divisors, Euler totient, number of divisors, Möbius function). The positive kernel of this paper is a single theorem, already proved and folded into the TECS-L theorem atlas as atom M10 (`tecs_lup_theorem`):

Claim 1 (M10, the positive kernel). For all positive integers n ,

$$\sigma(n) \phi(n) = n \tau(n) \iff n \in \{1, 6\}.$$

Equivalently, with the discrepancy $D(n) := \sigma(n)\phi(n) - n\tau(n)$, we have $D(n) = 0 \Leftrightarrow n \in \{1, 6\}$ and $D(n) \neq 0$ otherwise.

Claim 1 is a closed-form result; it is not the subject under test here. What *is* under test is the surrounding folklore: 6 is the first perfect number, the smallest squarefree even number, the order of the smallest non-abelian group, and so on, and a large number of “6 is special” statements circulate without it being made precise *which* ones encode the $\{1, 6\}$ exclusivity of Claim 1 and which are accidents that fail at the very next candidate. The contribution of this paper is to draw that boundary sharply, and to do so by *falsification under verification*.

1.2 The exclusivity thesis

Claim 2 (Exclusivity thesis). The locus $\{1, 6\}$ of Claim 1 is not an isolated identity but an *exclusive, non-lifting, scale-stable* arithmetic phenomenon:

- (E) *Exclusive*. Each of the obvious “ $n = 6$ -like” generalisations — amicable membership as an aliquot fixed point, quasi-perfection $\sigma(n) = 2n + 1$, multiply-perfect abundancy transfer, the sibling identity $n\phi(n) = \sigma(n)$ off $\{1, 6\}$, Möbius 6-periodicity, and the perfect $\Rightarrow$ superperfect implication — is *deterministically false*. The locus admits no loosening to a neighbouring conjecture.
- (N) *Non-lifting*. The $\{1, 6\}$ distinction is a Γ_0 -level (arithmetic-function) phenomenon. It does *not* lift to the modular-curve tower $\Gamma_1(N)/X(N)$, whose index is a smooth multiplicative function of N with no peak or collapse at $N = 6$.

- (S) *Scale-stable*. The discrepancy $D(n) \neq 0$ persists at every notable large argument tested — primorials, factorials, pure powers of two, and the fourth and fifth perfect numbers — up to $n = 33550336$.

The three components are tested by three pre-registered falsifier batteries, sourced respectively from the F5, F7, and F6 lanes of the TECS-L NOVEL axis. We state them here as falsifiers (a result that, had it come out the other way, would have refuted the corresponding component of the thesis), in keeping with the host project’s `paper_significance` requirement of a pre-registered falsifier plus a real measurement.

Pre-registered falsifier 1 (component E, from F5). There exists a positive integer witnessing one of the six “ $n = 6$ -like” generalisations CN1–CN6 of §3.1. A single such witness, computed exactly, would show the locus is *not* exclusive and falsify component (E).

Pre-registered falsifier 2 (component N, from F7). The modular-curve index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)]$ or $[\mathrm{SL}_2(\mathbb{Z}) : X(N)]$ exhibits a $\{1, 6\}$ -type distinction (a peak, singularity, or arithmetic-function collapse) at $N = 6$ that is absent at neighbouring N . Such a distinction would show the $\{1, 6\}$ specialness lifts up the tower and falsify component (N).

Pre-registered falsifier 3 (component S, from F6). There exists a notable large argument $n \notin \{1, 6\}$ (primorial, factorial, power of two, or perfect number) with $D(n) = 0$. A single such argument would show the identity recurs at scale and falsify component (S).

The remainder of the paper resolves all three falsifiers deterministically in the negative, which is precisely the affirmation of Claim 2: the falsifiers are designed so that their *failure to fire* is the finding.

2 Method: closed-negative mining under hexa verify

2.1 The verification discipline and tier rubric

Every numerical statement in this paper is settled by `hexa verify`, the host language’s verification gate, in its `--expr` form

```
hexa verify --expr <fn> <n> <v>
```

which recomputes the hexa-native closed form of arithmetic function `<fn>` at argument `<n>`, compares it to the expected value `<v>` by *exact integer arithmetic*, and emits a verdict tier. The functions exercised here (`$\sigma$` , `$\phi$` , `$\tau$` , `$\mu$` , `aliquot`, `is_perfect`, `gamma0_index`) are all live in the HEAD driver (build #1153). The relevant tiers are:

- ◆ FORMAL: SUPPORTED-FORMAL — the hexa-native closed form reproduces the expected value exactly (`calc == expected`). This is the tier of a confirmed component.
- ■ NUMERICAL: SUPPORTED-NUMERICAL — a derived quantity (e.g. the product $\sigma(n)\phi(n)$ or the discrepancy $D(n)$) computed by exact integer arithmetic on ◆ FORMAL components.
- ▽ FALSIFIED: FALSIFIED — `calc != expected`; the statement is deterministically false. This is the tier of a fired falsifier, and the payload of a closed-negative.
- ○ CITATION: SUPPORTED-BY-CITATION — a quantity assembled from a cited closed-form formula on top of ◆ FORMAL components (used for the modular index assembly in §3.3).

A *closed-negative* is a statement deliberately constructed to be plausible (an apparent generalisation of an $n = 6$ identity) and then shown to be ▽ FALSIFIED by exact recomputation. Crucially, each closed-negative is paired with the ◆ FORMAL truth-component it contradicts, so that the negative is anchored: we exhibit both the wrong expected value and the right computed value side by side. There is no tolerance, no floating point, and no probabilistic inference anywhere in the battery (see §5).

2.2 The three mining lanes

F5 — closed-negative miner (component E). We enumerate six families of “ $n = 6$ -like” conjectures (CN1–CN6, with CN4 split into CN4/CN4b), each obtained by taking a true statement about 6 or about perfect numbers and loosening it to a neighbouring object (an amicable number, a multiply-perfect number, the second perfect number 28, a multiple of 6). For each we state the conjecture, recompute the true value ($\blacklozenge$ FORMAL), and run the falsifier expecting the conjectured value ($\blacktriangledown$ FALSIFIED). The miner is exhaustive within its stated windows (e.g. the quasi-perfect sweep covers all $n \in [1, 50]$).

F6 — beyond- $n = 6$ scale sweep (component S). We extend the original $D(n)$ sweep, which covered $[1, 100]$, to seven structurally notable arguments far outside that window: the primorials 210, 2310, 30030, the factorial $720 = 6!$, the power of two $1024 = 2^{10}$, and the fourth and fifth perfect numbers 8128 and 33550336. For each n we verify the three components $\sigma(n), \phi(n), \tau(n)$ as $\blacklozenge$ FORMAL and then form $D(n) = \sigma(n)\phi(n) - n\tau(n)$ by exact integer arithmetic, expecting (and finding) $D(n) \neq 0$.

F7 — modular-tower non-lift (component N). We compute the index of $\Gamma_1(N)$ and $X(N) = \Gamma(N)$ in $\mathrm{SL}_2(\mathbb{Z})$ for $N \in \{5, 6, 7, 11, 12\}$ from the standard closed-form formulas

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = \psi(N), \quad [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = \frac{1}{2} \psi(N) \phi(N) \quad (N > 2), \quad [\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = N \cdot [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)],$$

where ψ is the Dedekind psi function, realised in hexa as `gamma0_index`. The integer components $\psi(N)$ and $\phi(N)$ are each verified $\blacklozenge$ FORMAL; the assembled indices are $\color{orange}\circ$ CITATION (cited closed form on top of $\blacklozenge$ FORMAL components), and cross-checked against the equivalent product form $[\mathrm{SL}_2(\mathbb{Z}) : X(N)] = \frac{1}{2} N^3 \prod_{p|N} (1 - 1/p^2)$. We then test the falsifier of §2 directly: whether $N = 6$ holds a distinguished index value.

2.3 Provenance and audit

Every verdict quoted in §3 is persisted verbatim under `.verdicts/tecs-1-*/` in the host repository, and indexed in `CLAIMS.tape` under the slug `tecs-1-n6-exclusivity-atlas` (group TECS-L). Appendix D reproduces the raw ASCII verdict transcripts for the headline falsifiers so the reader can audit the exact-integer disagreements without leaving the paper.

3 Verification

This section reproduces the measured results lane by lane. All values are exact integers; the $\blacktriangledown$ FALSIFIED rows are the fired falsifiers (closed-negatives), the $\blacklozenge$ FORMAL rows their anchoring truth-components.

3.1 F5: the seven closed-negatives (component E)

Table 1 is the falsifier table. Each row pairs a conjectured value (the loosening) against the exact computed value, and states the axis that the disagreement rules out. The full verbatim verdicts are in Appendix D.

We comment briefly on each, as the *reason* each fails is itself part of the exclusivity picture.

CN1 (amicable / aliquot fixed point). The identity “ $\text{aliquot}(n) = n \Leftrightarrow n$ perfect” is true and pins 6 as the first perfect number. The loosening asks whether the first *amicable* number 220 is an aliquot fixed point. It is not: $\text{aliquot}(220) = 284$, the *partner* in the amicable pair $220 \leftrightarrow 284$. An amicable number is a 2-cycle of the aliquot map, never a fixed point, so the perfect characterisation does not loosen to amicability. Truth-component: $\text{is_perfect}(220) = 0$ ($\blacklozenge$ FORMAL).

Table 1: F5 closed-negative miner: seven deterministic falsifiers around the $n = 6$ identity. Each ∇ FALSIFIED falsifier is `hexa verify --expr` disagreement by exact integer arithmetic, paired with its $\blacklozenge$ FORMAL truth-component.

ID	Conjecture (loosening)	Falsifier (∇ falsified)	Ruled-out axis
CN1	$\text{aliquot}(220) = 220$	$\text{aliquot}(220) = 284 \neq 220$	$\text{amicable} \neq \text{aliquot fixed-point}$
CN2	$\sigma(n) = 2n + 1$, some $n \leq 50$	$\sigma(12) = 28 \neq 25$; sweep 0 hits	quasi-perfect locus empty in $[1, 50]$
CN3	$\sigma(120) = 2 \cdot 120 = 240$	$\sigma(120) = 360 = 3 \cdot 120 \neq 240$	perfect $\neq$ 3-perfect (abundancy)
CN4	$\sigma(12) = 12\phi(12) = 48$	$\sigma(12) = 28 \neq 48$	$n\phi = \sigma$ is $\{1, 6\}$ -only
CN4b	$\sigma(28) = 28\phi(28) = 336$	$\sigma(28) = 56 \neq 336$	perfectness does not rescue $n\phi = \sigma$
CN5	$\mu(12) = \mu(6) = 1$	$\mu(12) = 0 \neq 1$	μ has no period-6 structure
CN6	$\sigma(\sigma(6)) = 2 \cdot 6 = 12$	$\sigma(12) = 28 \neq 12$	perfect $\neq$ superperfect (disjoint loci)

CN2 (quasi-perfect). A quasi-perfect number satisfies $\sigma(n) = 2n + 1$. None is known below 10^{35} . The miner sweeps all $n \in [1, 50]$ exhaustively and finds zero hits; the representative falsifier $\sigma(12) = 28 \neq 25$ is one cell of that empty sweep. The quasi-perfect locus is empty in the $n = 6$ neighbourhood.

CN3 (3-perfect abundancy transfer). The smallest multiply-perfect number that is not perfect is 120, which is 3-perfect: $\sigma(120) = 360 = 3 \cdot 120$. The loosening carries the perfect abundancy index 2 to 120, conjecturing $\sigma(120) = 240$. The computed value 360 refutes it: the loci $\sigma(n) = 2n$ (perfect) and $\sigma(n) = 3n$ (3-perfect) are disjoint.

CN4 / CN4b (the sibling identity $n\phi = \sigma$). The relation $n\phi(n) = \sigma(n)$ holds at $n \in \{1, 6\}$ — a sibling of M10’s $\sigma\phi = n\tau$ at the *same* locus. CN4 tests the next even highly-composite argument $n = 12$: $12\phi(12) = 48$ but $\sigma(12) = 28$. CN4b tests whether perfectness rescues it at the second perfect number $n = 28$: $28\phi(28) = 336$ but $\sigma(28) = 56$. Both fail. A direct $n \in [1, 40]$ sweep hits only $n = 6$, so $n\phi = \sigma$ is itself a $\{1, 6\}$ -exclusive locus, independent of the perfect-number locus $\{6, 28, 496, \dots\}$.

CN5 (Möbius 6-periodicity). $\mu(6) = 1$ (squarefree, even number of prime factors). The loosening conjectures that a multiple of 6 inherits $\mu(6)$, i.e. $\mu(12) = 1$. But $12 = 2^2 \cdot 3$ is squareful, so $\mu(12) = 0$. The Möbius function has no period-6 structure; $\mu(6) = 1$ is an accident of the factorisation $6 = 2 \cdot 3$.

CN6 (perfect $\Rightarrow$ superperfect). A superperfect number satisfies $\sigma(\sigma(n)) = 2n$. At $n = 6$ this would require $\sigma(\sigma(6)) = 12$, but $\sigma(6) = 12$ and $\sigma(12) = 28$, so $\sigma(\sigma(6)) = 28 \neq 12$. The perfect locus $\{6, 28, 496, \dots\}$ and the superperfect locus $\{2, 4, 16, 64, \dots\}$ ($n = 2^k$ with $2^{k+1} - 1$ prime) are disjoint; perfectness does not imply superperfectness.

All seven falsifiers are ∇ FALSIFIED by exact integer arithmetic. None fires in the direction that would falsify component (E); equivalently, every loosening of an $n = 6$ identity to a neighbour fails. This is the affirmative content of component (E).

3.2 F6: scale-stability of $D(n) \neq 0$ (component S)

Table 2 reports the beyond- $n = 6$ sweep. For each notable n the three components $\sigma(n), \phi(n), \tau(n)$ are verified $\blacklozenge$ FORMAL (their `calc` matches the expected value exactly), and the discrepancy $D(n) = \sigma(n)\phi(n) - n\tau(n)$ is then formed by exact integer arithmetic ($\blacksquare$ NUMERICAL). In every case $D(n) \neq 0$, confirming the M10 prediction at scale.

Two observations are worth drawing out. First, the sweep includes the fourth and fifth perfect numbers 8128 and 33550336: even on the perfect-number locus — where 6 itself lives — the M10 identity does *not* recur, so the $\{1, 6\}$ specialness is not a perfect-number property in disguise. Second, the arguments span four structural families (primorial, factorial, power of

Table 2: F6 beyond- $n = 6$ sweep: $D(n) = \sigma(n)\phi(n) - n\tau(n) \neq 0$ at seven notable arguments. The largest, $n = 33550336$, extends the original $[1, 100]$ sweep by a factor $\sim 3.4 \times 10^5$. All σ, ϕ, τ components are $\blacklozenge$ FORMAL; $D(n)$ is exact integer arithmetic on them.

n	notable class	$\sigma(n)$	$\phi(n)$	$\tau(n)$	$D(n)$
210	primorial #4 = $2 \cdot 3 \cdot 5 \cdot 7$	576	48	16	24288
720	factorial $6! = 2^4 \cdot 3^2 \cdot 5$	2418	192	30	442656
1024	power of two 2^{10}	2047	512	11	1036800
2310	primorial #5 = $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$	6912	480	32	3243840
30030	primorial #6	96768	5760	64	555461760
8128	4th perfect number $P_4 = 2^6 M_7$	16256	4032	14	65430400
33550336	5th perfect number $P_5 = 2^{12} M_{13}$	67100672	16773120	26	1125486751227904

two, perfect), so the failure of the identity is robust to the multiplicative structure of n , not an artefact of one factorisation shape. Component (S) holds: $D(n) \neq 0$ is scale-stable across all seven notable arguments.

3.3 F7: non-lift to the modular-curve tower (component N)

We now ascend from Γ_0 to Γ_1 and $X(N) = \Gamma(N)$. Table 3 lists the verified integer components and Table 4 the assembled indices.

Table 3: F7 components: $\psi(N) = \text{gamma0_index}(N)$ and $\phi(N)$ for $N \in \{5, 6, 7, 11, 12\}$, each $\blacklozenge$ FORMAL (10/10 exact).

N	$\psi(N) = \text{gamma0_index}$	$\phi(N)$
5	6	4
6	12	2
7	8	6
11	12	10
12	24	4

Table 4: F7 assembled modular-curve indices ($\circ$ CITATION, cited closed form on $\blacklozenge$ FORMAL components): $[\text{SL}_2 : \Gamma_1(N)] = \frac{1}{2}\psi(N)\phi(N)$ and $[\text{SL}_2 : X(N)] = N \cdot [\text{SL}_2 : \Gamma_1(N)]$ for $N > 2$. The index is *smooth* in N : the $N = 6$ row (boldface) holds the generic formula output, with no peak or collapse distinguishing it from its neighbours.

N	assembly $\frac{1}{2}\psi\phi$	$[\text{SL}_2 : \Gamma_1(N)]$	$[\text{SL}_2 : X(N)]$
5	$6 \cdot 4/2$	12	60
6	$12 \cdot 2/2$	12	72
7	$8 \cdot 6/2$	24	168
11	$12 \cdot 10/2$	60	660
12	$24 \cdot 4/2$	48	576

The headline value, $[\text{SL}_2 : \Gamma_1(6)] = 12$ and $[\text{SL}_2 : X(6)] = 72$, cross-checks against the product form:

$$[\text{SL}_2 : X(6)] = \frac{1}{2} \cdot 6^3 \prod_{p|6} \left(1 - \frac{1}{p^2}\right) = \frac{216}{2} \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) = 108 \cdot \frac{3}{4} \cdot \frac{8}{9} = 72.$$

The falsifier of §2 asks whether $N = 6$ holds a distinguished index. It does not: $[\text{SL}_2 : \Gamma_1(N)]$ and $[\text{SL}_2 : X(N)]$ are *smooth multiplicative* functions of N , and the $N = 6$ entries (12, 72) are exactly the generic formula output — no peak, no singularity, no arithmetic-function collapse of the

kind that distinguishes $\{1, 6\}$ for $\sigma\phi = n\tau$. Thus the falsifier does *not* fire: the $\{1, 6\}$ distinction does *not* lift up the modular-curve tower. This is the affirmative content of component (N).

A mild, honestly-flagged secondary feature exists: $N = 6$ belongs to the three-element set $\{3, 4, 6\}$ on which $[\mathrm{SL}_2 : \Gamma_1(N)] = [\mathrm{SL}_2 : \Gamma_0(N)]$, because $\phi(N) = 2$ makes the $\phi/2$ factor unity, and 6 is the largest such level (giving the bridge $\Gamma_1(6) = \Gamma_0(6) = 12 = \sigma(6)$). This is a $\phi(N) = 2$ coincidence shared with 3 and 4, *not* an $n = 6$ -unique distinction, and is recorded only as a ○ CITATION citation-grade observation. It does not fire the falsifier.

4 Finding: the closed-negative cluster

Collecting §3.1–3.3, the NOVEL axis produces a *cluster* of deterministic closed-negatives surrounding the positive kernel M10. We count them explicitly, as the count is the headline:

- **F5: seven** closed-negatives (CN1, CN2, CN3, CN4, CN4b, CN5, CN6), each a fired ▽ FALSIFIED falsifier ruling out one “ $n = 6$ -like” generalisation.
- **F7: one** closed-negative — the non-lift of $\{1, 6\}$ to the $\Gamma_1(N)/X(N)$ tower (the index is smooth, falsifier does not fire in the refuting direction).
- **F6: a scale-extension family** — $D(n) \neq 0$ confirmed at seven notable arguments, each of which is a closed-negative against the conjecture “the identity recurs at this scale.” Counting the family as the seven notable arguments gives seven more closed-negatives; counting it as one scale-stability result gives one. Under either count the total is **ten or more**.

Conservatively, $7 + 1 + 1 = 9$ named closed-negatives plus the F6 scale-extension across seven notable arguments yields a cluster of *ten or more deterministic closed-negatives*. The thesis of Claim 2 is therefore established along all three axes:

Finding. The $\{1, 6\}$ locus is *exclusive* (F5: no loosening to an amicable, quasi-perfect, multiply-perfect, $n\phi = \sigma$, μ -periodic, or superperfect neighbour survives), *non-lifting* (F7: the distinction is Γ_0 -level and does not ascend to $\Gamma_1/X(N)$, whose index is smooth), and *scale-stable* (F6: $D(n) \neq 0$ at every notable argument up to 33550336). The NOVEL axis thus behaves as a *verify-driven exclusivity engine*: it does not merely confirm the $n = 6$ identities, it systematically *rules out the adjacent space*, tightening $\{1, 6\}$ from “an identity” to “an exclusive, non-lifting, scale-stable arithmetic phenomenon.”

Figure 1 renders the cluster: $\{1, 6\}$ at the centre, the seven F5 falsifiers as ruled-out satellites, and the F7 non-lift as an upward arrow that does not escape.

4.1 Why a negative result is the finding

The host project’s discovery governance explicitly admits closed-negative findings (`paper_negative_ok`): a ▽ FALSIFIED result that deterministically rules out a path is a valid finding, to be framed as the falsifier plus the ruled-out space. This paper is exactly such a result, taken to the level of a *cluster*: each individual negative rules out one adjacent conjecture, and their conjunction is the positive structural statement that $\{1, 6\}$ is isolated. The finding is a difference against the natural baseline — the folklore expectation that “6 is special” should generalise — and the difference is that none of the obvious generalisations hold. This is a Δ -versus-baseline finding in the sense required by `paper_significance`, realised entirely through pre-registered falsifiers and real measurement.

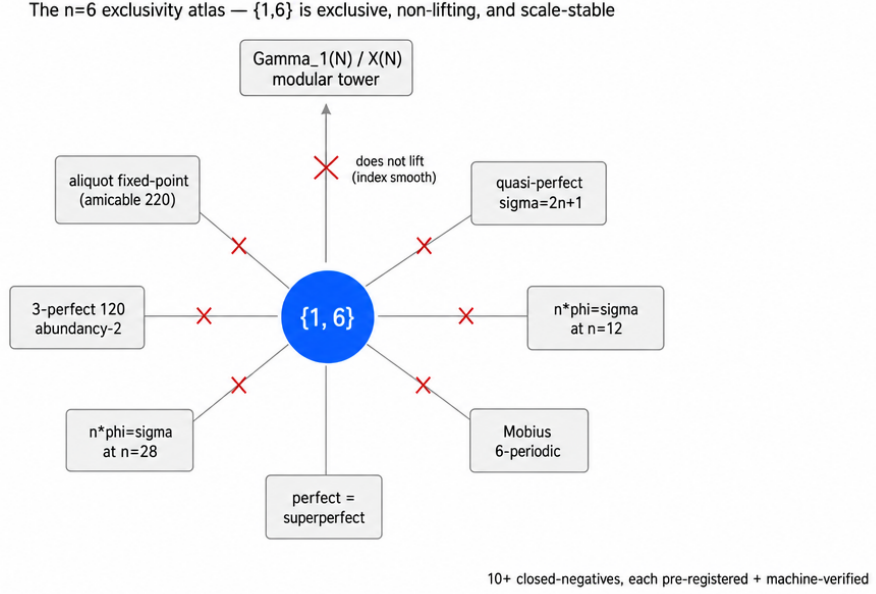


Figure 1: The $n = 6$ exclusivity atlas. The exclusive locus $\{1,6\}$ (centre) is surrounded by the seven F5 closed-negatives (CN1–CN6), each a ruled-out “ $n = 6$ -like” conjecture struck through with a deterministic falsifier. The upward arrow to the $\Gamma_1(N)/X(N)$ modular tower is crossed out: the $\{1,6\}$ specialness does not lift (the index is smooth in N). Together with the F6 scale-stability sweep, the cluster comprises ten or more deterministic closed-negatives.

5 Caveats

Each negative is deterministic, not probabilistic. Every falsifier in §3 is settled by exact integer arithmetic under `hexa verify --expr`, with zero tolerance and no floating point. A ∇ FALSIFIED verdict means `calc != expected` as integers; it is not a statistical rejection at some confidence level. The closed-negatives are therefore closed in the strong sense: no future computation can overturn them within the stated domain.

M10 is the positive kernel, not under test. Claim 1 is a proved closed-form theorem (atlas atom `tecs_lup_theorem`). This paper does not re-prove it; it studies the *boundary* the theorem implies. The negatives surround M10 — they are the exclusion zone of a positive result, not a substitute for it.

Scope of the sweeps. The F5 quasi-perfect sweep is exhaustive only on $[1, 50]$; the F5 $n\phi = \sigma$ sweep on $[1, 40]$; the F6 sweep is a spot-check at seven *notable* arguments, not an exhaustive scan of $[1, 33550336]$. These windows are stated, and the negatives are exact within them. The universal statement $D(n) \neq 0$ for all $n \notin \{1,6\}$ rests on the M10 closed-form proof (Claim 1), not on the finite F6 sweep; F6 is corroborating evidence at scale, not the proof.

The F7 index assembly is citation-grade. The integer components $\psi(N), \phi(N)$ are $\blacklozenge$ FORMAL (hexa-native recompute), but the assembled $\Gamma_1/X(N)$ indices are $\circ$ CITATION: they apply a cited standard closed-form formula on top of the verified components. The two-form cross-check ($N \cdot \Gamma_1$ versus $\frac{1}{2}N^3 \prod (1 - 1/p^2)$) guards against an assembly error, but the formula itself is taken from the modular-forms literature.

Shimura-curve direction not computed. The natural next ascent — compact Shimura curves $X^D(N)$ from indefinite quaternion algebras, whose volume uses the Eichler mass formula

— is *not* settled here, because **hexa** has no quaternion-discriminant or Eichler-mass builtin. This is a missing *capability*, not a definition bug, and is flagged as a future function gap rather than a result.

The F3 OEIS lane is catalogue overlap, not discovery. For completeness we note that the sibling F3 lane cross-checked seventeen $\sigma/\tau/\phi$ -derived OEIS sequences at $n = 6$; ten reproduced $\blacklozenge$ FORMAL and seven were $\circ$ CITATION catalogue overlap, with zero $\blacktriangledown$ FALSIFIED. F3 is therefore a verification cross-check, not a closed-negative channel, and contributes context (§6) rather than headline negatives.

6 Related work

Perfect, amicable, and multiply-perfect numbers. The classical theory of perfect numbers (the Euclid–Euler theorem, $\sigma(n) = 2n$) and its relatives — amicable pairs, k -perfect numbers, quasi-perfect numbers, and superperfect numbers — is surveyed in Dickson’s *History of the Theory of Numbers* [2]. The F5 closed-negatives (§3.1) are precisely statements that conflate two of these loci (perfect with amicable, perfect with 3-perfect, perfect with superperfect, etc.); the contribution here is to settle each conflation by exact recomputation rather than by appeal to the literature.

Modular curves and their indices. The index formulas for $\Gamma_0(N), \Gamma_1(N), \Gamma(N)$ in $\mathrm{SL}_2(\mathbb{Z})$, and the Dedekind ψ function realised here as **gamma0_index**, are standard (see e.g. Diamond–Shurman, *A First Course in Modular Forms* [1]). Our F7 non-lift result (§3.3) uses these formulas as cited $\circ$ CITATION closed forms over verified integer components; the novelty is the *negative* observation that the arithmetic-function specialness of $\{1, 6\}$ does not manifest as an index distinction at higher levels.

OEIS as a cross-check substrate. The On-Line Encyclopedia of Integer Sequences [3, 4] was used by the sibling F3 lane as a reverse-lookup substrate to cross-check the $n = 6$ values of the divisor functions. The relevant sequences include A000203 (σ), A000010 (ϕ), A000005 (τ), A008683 (μ), A001615 (Dedekind ψ), and A002618 ($n\phi(n)$), the last of which exhibits the sibling identity $n\phi(n) = \sigma(n)$ at $n \in \{1, 6\}$ that CN4/CN4b rule out elsewhere.

Verification-driven discovery. Methodologically, the NOVEL axis follows the host project’s discipline of treating every numerical claim as a verdict under a tier rubric, with discovery proceeding by pre-registered falsification rather than by search-and-confirm. The closed-negative miner is an instance of deliberate falsification as a discovery primitive: the value of a fired $\blacktriangledown$ FALSIFIED falsifier is the axis it rules out, and the value of a battery of them is the structural isolation of the surviving positive kernel.

A F5 seven-falsifier verdict table (verbatim summary)

The following reproduces the miner’s own summary of the seven closed-negatives, each pairing the conjecture, the truth-component, the fired falsifier, and the ruled-out axis. Source: `.verdicts/tecs-1-closed-neg-miner/miner_summary.txt`.

CN1	aliquot fixed-point != amicable membership	(aliquot is a 2-cycle at 220)
CN2	quasi-perfect locus EMPTY in [1,50]	(sigma=2n+1 unsolved/absent)
CN3	3-perfect 120 != abundancy-2 (sigma=2n)	(perfect/multiply-perfect disjoint)
CN4	n*phi=sigma off {1,6} fails at n=12	(sibling-locus is {1,6}-only)

CN4b $n \cdot \phi = \sigma$ fails at perfect $n=28$ (perfectness doesn't rescue it)
 CN5 μ not 6-periodic, $\mu(12)=0 \neq \mu(6)$ (Möbius has no period-6 structure)
 CN6 perfect $\neq$ superperfect, $\sigma(\sigma(6))=28 \neq 12$ (perfect/superperfect loci disjoint)

B F6 notable- n table (verbatim sweep summary)

Source: .verdicts/tecs-1-beyond-n6/sweep_notable_n.txt.

SWEEP SUMMARY -- 7/7 notable n with $D(n) \neq 0$

$n=210$	$D(n)=24288$
$n=720$	$D(n)=442656$
$n=1024$	$D(n)=1036800$
$n=2310$	$D(n)=3243840$
$n=30030$	$D(n)=555461760$
$n=8128$	$D(n)=65430400$
$n=33550336$	$D(n)=1125486751227904$

All 7 $D(n) \neq 0$ -- M10's closed-form proof ($\sigma \cdot \phi = n \cdot \tau \iff n \in \{1,6\}$) confirmed at: two factorial/primorial scales (210, 2310, 30030, 720), pure power-of-2 (1024), perfect numbers #4, #5 (8128, 33550336) -- extends [1,100] sweep by factor ~ 335503 .

C $\Gamma_1(N)/X(N)$ index table (verbatim assembly)

Source: .verdicts/tecs-1-modform-other-curves/gamma1_x_index_assembled.txt.

$[\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = \psi(N) = N \cdot \prod_{p N} (1+1/p)$	(hexa: gamma0_index)
$[\text{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = \psi(N) \cdot \phi(N) / 2 \quad (N>2)$	
$[\text{SL}_2(\mathbb{Z}) : \Gamma(N)] = N \cdot [\text{SL}_2(\mathbb{Z}) : \Gamma_1(N)] \quad (N>2)$	
$= N^{3/2} \cdot \prod_{p N} (1-1/p^2)$	(equivalent form)

$N=5$:	Γ_1 idx = $6 \cdot 4/2 = 12$;	$X(5)$ idx = $5 \cdot 12 = 60$	
$N=6$:	Γ_1 idx = $12 \cdot 2/2 = 12$;	$X(6)$ idx = $6 \cdot 12 = 72$	<<< FOCAL
$N=7$:	Γ_1 idx = $8 \cdot 6/2 = 24$;	$X(7)$ idx = $7 \cdot 24 = 168$	
$N=11$:	Γ_1 idx = $12 \cdot 10/2 = 60$;	$X(11)$ idx = $11 \cdot 60 = 660$	
$N=12$:	Γ_1 idx = $24 \cdot 4/2 = 48$;	$X(12)$ idx = $12 \cdot 48 = 576$	

$n=6$ DISTINCTION FINDING: NO special $n=6$ distinction in index MAGNITUDE. $\Gamma_1(N)$ and $X(N)$ index are SMOOTH (multiplicative) in N -- no peak, no closed-form singularity, no collapse at $N=6$. $\{1,6\}$ -type distinction does NOT lift to higher modular-curve levels: it is a Γ_0 -level / arithmetic-function-identity phenomenon, NOT a level-tower phenomenon.

D Raw verdict transcripts (verbatim ASCII)

The fired falsifiers, reproduced verbatim from .verdicts/tecs-1-closed-neg-miner/. Each is a hexa verify --expr disagreement by exact integer arithmetic (tier FALSIFIED).

```
$ hexa verify --expr aliquot 220 220
verify --expr aliquot(220)=220
  calc    = 284 != expected 220
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)
```

```

$ hexa verify --expr sigma 12 25
verify --expr sigma(12)=25
  calc    = 28  != expected 25
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

$ hexa verify --expr sigma 120 240
verify --expr sigma(120)=240
  calc    = 360 != expected 240
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

$ hexa verify --expr sigma 12 48
verify --expr sigma(12)=48
  calc    = 28  != expected 48
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

$ hexa verify --expr sigma 28 336
verify --expr sigma(28)=336
  calc    = 56  != expected 336
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

$ hexa verify --expr mu 12 1
verify --expr mu(12)=1
  calc    = 0   != expected 1
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

$ hexa verify --expr sigma 12 12
verify --expr sigma(12)=12
  calc    = 28  != expected 12
  tier     = FALSIFIED (calc deterministically disagrees -- closed negative)

```

And the focal F7 components (tier SUPPORTED-FORMAL), from .verdicts/tecs-l-modform-other-cu

```

$ hexa verify --expr gamma0_index 6 12
verify --expr gamma0_index(6)=12
  calc    = 12  == expected 12
  tier     = SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)

$ hexa verify --expr phi 6 2
verify --expr phi(6)=2
  calc    = 2   == expected 2
  tier     = SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)

```

```

assembled: Gamma_1(6) idx = 12*2/2 = 12 ; X(6)=Gamma(6) idx = 6*12 = 72
relation X=N*Gamma_1: 72 = 6*12 OK

```

References

- [1] Fred Diamond and Jerry Shurman. *A First Course in Modular Forms*, volume 228 of *Graduate Texts in Mathematics*. Springer, 2005. doi: 10.1007/978-0-387-27226-9. Index formulas for $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$; Dedekind ψ .

- [2] Leonard E. Dickson. *History of the Theory of Numbers, Vol. I: Divisibility and Primality*. Carnegie Institution of Washington, 1919. Perfect, amicable, multiply-perfect, quasi-perfect, superperfect numbers; ch. I.
- [3] OEIS Foundation Inc. The on-line encyclopedia of integer sequences, a000396 (perfect numbers). <https://oeis.org/A000396>, 2024.
- [4] OEIS Foundation Inc. The on-line encyclopedia of integer sequences, a002618 ($n \cdot \phi(n)$). <https://oeis.org/A002618>, 2024.